MIL-A-15344J(NAVY)
7 June 1963
SUPERSEDING
MIL-A-15344H(NAVY)
14 November 1961

MILITARY SPECIFICATION

AMPLIFIER, SYNCHRO SIGNAL

All interested bureaus of the Department of the Navy have concurred in the use of this specification.

1. SCOPE

1.1 Scope.- This specification covers the requirements for synchro signal amplifiers for the purpose of providing a means of increasing the capacity of a synchro signal transmission system, converting from 60 to 400 cycle synchro signal transmission or from 400 to 600 cycle synchro signal transmission, and to isolate reflected oscillations from the input synchro signal bus.

1.2 Classification.- The synchro signal amplifier shall be of the types listed in table I, as specified (see 6.2).

Table I - Synchro signal amplifier.

Type	Synchro excitation frequency cycles per second (c.p.s.).		Synchro units							
			Input				Output			
	Input	Output	1-speed	18-speed	36-speed	648-speed	1-speed	18-speed	36-speed	648-speed
E	60	60 and 400	18CT6 or 23CT6	- - -	18CT6 or 23CT6	- - -	6G and 31TX4	- - -	6G and 31TX4	- - -
E	60	60 or 400 (not both)	18CT6 or 23CT6	18CT6 or 23CT6	- - -	18CT6 or 23CT6	23TX4	23TX4	- - -	23TX4
F	400	60 and 400	18CT4 or 23CT4	- - -	18CT4 or 23CT4	- - -	6G and 31TX4	- - -	6G and 31TX4	- - -

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-I-983 - *Interior Communication Equipment, Naval Shipboard; Basic Design Requirements for.*
MIL-S-2335 - Synchros, 60 Cycle, 115 Volt.
MIL-Q-9858 - Quality Control System Requirements.
MIL-S-17087 - Servo Motors, 2 Phase, 400 Cycles.
MIL-E-17555 - Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of.
MIL-S-20708 - Synchros, 60 and 400 Cycle, General Specification.

STANDARDS

MILITARY

MIL-STD-710 - Synchros, 60 and 400 Cycle.

FSC 5990

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DRAWINGS

BUREAU OF SHIPS
9000-S6202-73907 - Switchboard, Indicator Light - 2, Lamp - SPF- Type B27A.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply:

OFFICIAL CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules.

(Application for copies should be addressed to Official Classification Committee, 1 Park Avenue at 33rd St., New York 16, New York.)

3. REQUIREMENTS

3.1 Qualification.- The synchro signal amplifier equipment furnished under this specification shall be a product which has been tested, and passed the qualification tests specified herein, and has been listed on or approved for listing on the applicable qualified products list.

3.2 Description.- The synchro signal amplifier shall consist of an amplifier, synchro control transformers (see table I); one alternating current (a.c.) two phase servo motor with necessary gear train for driving the synchro rotors, as specified in 3.7.2; and output synchro signal generators of appropriate size as specified (see 6.2) all assembled in a dripproof enclosure (see 3.6.1) with the necessary operating controls (see 3.6.4), indicator lights (see 3.7.4), alarm relays (see 3.7.5) and capacitors (see 3.7.6).

3.3 Operation.- The typical cycle of operation of the synchro signal amplifier is as follows: A synchro signal, such as that obtained from a master gyrocompass is transmitted to the synchro control transformers (CTs) in the synchro amplifier. The signal received by the synchro control transformers is fed to the amplifier unit where it is intensified and used to actuate the follow-up motor which, through gearing, turns the synchro control transformer rotors until the induced voltages are reduced to a null, thereby stopping the follow-up action. At the same time the follow-up motor also drives, through gearing, the output synchro signal generator rotors. The output synchro signal generators actuate the remote network of synchro signal receivers which are then displaced the same angular amount as the input synchro signal.

3.4 General features.- In addition to the requirements specified herein, the equipment shall be in accordance with MIL-I-983 except for the following paragraphs. (Whenever a requirement of MIL-I-983 conflicts with a requirement of this specification, the requirement of this specification shall govern):

Universal mounting
Stand mounting
Panel mounting
Bulkhead mounting
Tumbling
Gunblast
Depth charge
Dial illumination
Repair parts (electronic)
Salt spray

3.5 Design.- The synchro signal amplifier shall be so designed that vibration, shock, or acceleration, which may be encountered in service aboard Naval ships, will not adversely affect the accuracy of the equipment, nor derange any part of the equipment beyond normal adjustment at sea. All parts except those necessary in electromagnetic circuits, ball bearings, shafts, thrust plates, and similar application where strength or hardness is essential shall be constructed of nonferrous, nonmagnetic materials.

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3.6 Mechanical features.-

3.6.1 Enclosure.- All parts required for the proper operation of the synchro signal amplifier shall be mounted in an enclosing case (degree of enclosure dripproof) designed for bulkhead, platform, and stack mounting (maximum of four units mounted one on top of another). Provision shall be made for the protection of gears from dust. Mounting and external wiring of the equipment shall be accomplished, insofar as practical, without having to remove the amplifier or synchro unit from the enclosure and without having to expose the gears to dust. A removable plate (a minimum of 4 inches by 8 inches by 1/8 inch) for the installation of stuffing tubes shall be provided on both sides of each enclosure. The enclosing case shall consist of an aluminum angle or channel framework, suitably welded to form a structure of adequate strength to support all subassemblies. A wrap-around or similar covering of aluminum shall be provided around the framework. Internal subassemblies shall not be supported by the aluminum covering. The case shall be sufficiently ventilated to meet the operating temperature requirements herein. Ventilation may be achieved by louvers in the ends and front cover, by screened openings in the bottom of the case and by umbrella type covered openings in the top.

3.6.2 Accessibility.- All units and subassemblies of the equipment shall be easily accessible for repair and adjustment. The access door to the amplifier shall be equipped with hinges and supporting straps or anchors and supporting chains so that it may act as a service platform for the amplifier internal assemblies. Protection of the internal subassemblies shall be provided in accordance with MIL-I-983. Supporting rods shall be furnished on the amplifier and servo units to allow repair work without the units resting on electrical components, electrical connections or on components that may be disarranged or otherwise damaged. A schematic diagram of the amplifier shall be attached to the inside of this door. Knurled knob captive screws with a screwdriver slot shall be used on all covers that require removal or opening for operating adjustments.

3.6.3 Weight and size.- The weight of a synchro signal amplifier shall be kept to a minimum and in no case shall the maximum weight of the various types exceed that specified in table II. Maximum size shall not exceed 3 cubic feet for any type of synchro amplifier.

Table II - Weights.

Type	Maximum weight with full complement of largest synchros required by this specification
	Pounds
E, F	120 when structureborne noise limits are not imposed.
E, F	130 when structureborne noise limits are imposed.

3.6.4 Controls and indicators.- The following operating controls and indicator lights shall be provided on the front of the enclosure:

(a) Alarm relay switch (see 3.7.5).
(b) Power and follow-up indicator lights (see 3.7.4).
(c) Fuseholders.

3.6.4.1 An indicator dial(s) (see 3.7.15) shall be visible through the front of the enclosure.

3.7 Electrical features.-

3.7.1 Amplifier unit.- The amplifier unit shall be electron tube, transistor or magnetic type and shall consist of the necessary items arranged on a removable chassis to provide control of the a.c. two-phase motor. Each line supplying power to the amplifier unit and the servo motor (see 3.7.3) shall be properly fused within the synchro amplifier. The total power required shall not exceed 150 watts. The amplifier

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chassis shall be grounded to the case, and a B minus bus isolated from the chassis shall be provided for the servo amplifier and shall be positively connected (bonded) to the case of the synchro amplifier at one physical point only. An isolation transformer or equivalent isolation shall be used between the output power of the servo amplifier and the servo motor control field.

3.7.2 Synchro unit.- The synchro unit shall consist of the input synchro control transformers, a two-phase servo motor, gear trains, and the output synchro signal generators to provide for the receipt of the input synchro signal from the master gyrocompass, or other synchro signal source, and the retransmission of the synchro signal to a network of synchro signal receivers. The plate supporting the synchro control transformers, servo motor, and synchro signal generators shall be so arranged as to be removable. All synchro transmitters and control transformers shall be contractor furnished material in accordance with MIL-S-20708 or MIL-S-2335, if for size 6 or 7 synchros, and in accordance with MIL-STD-710.

3.7.3 Servo motor.- The servo motor shall be a two-phase low-inertia design with grease sealed ball bearings, and shall be contractor furnished. The choice of servo-motor shall be approved by the bureau or agency concerned. Specification MIL-S-17087 shall apply for 400-cycle servo motors. Sixty cycle servo motors shall have temperature ratings comparable to those for 400-cycle servo motors.

3.7.4 Indicator lights.- Indicator lights having white lenses shall be provided to give a single indication of failure in (a) servo system follow-up action, (b) input power, (c) output 60-cycle synchro excitation power and (d) output 400-cycle synchro excitation power. An illuminated light shall indicate normal operation; and extinguished light - failure. The output synchro excitation lights shall be omitted and the openings in the front of the enclosure shall be concealed with a cover, or by other means when synchro differential units are substituted for synchro generators. Indicator lights shall be in accordance with Drawing 9000-S6202-73907.

3.7.5 Alarm relay and switch.- An alarm relay shall be provided to indicate failure of the servo system to follow the input synchro signal, loss of power to the servo amplifier and loss of output 60 and 400-cycle synchro power. The relay contacts shall be designed to carry 5 amperes at 0.5 power factor, 115 volts 60 or 400 cycles a.c., and shall be connected to terminal connections for an external alarm circuit. The relay shall operate when the servo error exceeds approximately 2.5 degrees or upon loss of input power or upon loss of output synchro excitation power. An alarm relay switch shall be provided, and shall be connected so that the alarm relay will operate the external alarm circuit as specified in table III. An example of an acceptable circuit is shown on figure 1. When only differential synchros constitute the 60-cycle or 400-cycle output synchros, the alarm circuitry and relays normally used for 60-cycle or 400-cycle output synchro generator power failure shall be omitted.

Table III - Alarm relay and switch connection.

Nominal values of follow-up error	Input and output synchro excitation	Alarm switch	External alarm circuit
------	Off	Off	De-energized
2.5° or less	On	Off	Energized
2.5° or less	On	On	De-energized
------	Off	On	Energized

3.7.6 Synchro capacitors.- The correct synchro capacitors shall be provided by the contractor (mounted within the enclosure) for all synchro control transformers. The value of the capacitor elements shall be determined by the contractor for the particular synchros to be used. The capacitor elements shall be rated for 600 volts d.c. or greater. Each synchro capacitor shall have the capacitance of its three capacitor elements balanced within 1 percent as measured by the divergence of the capacitance of any one leg from the arithmetical mean of the three.

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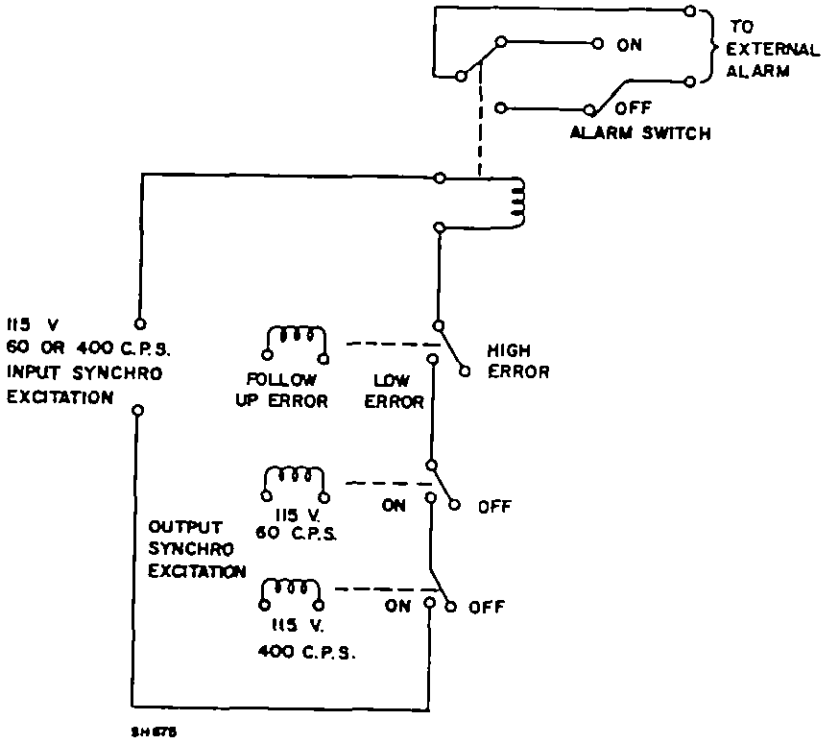


Figure 1 - Alarm circuit.

3.7.7 Input operation.- Synchro signal amplifiers to operate at 1 and 36-speed shall be supplied unless otherwise specified (see 6.2). The design shall provide for operation with the following inputs when simple gear substitutions or simple wiring changes are made:

For 1 and 36-speed units	For 1, 18 and 648-speed units
One and 36-speed.	One, 18 and 648-speed.
One speed.	One speed.
Thirty-six speed.	One and 18-speed.
Two speed.	Eighteen speed.
Two and 36-speed.	Eighteen and 648-speed.
	Six hundred and forty eight speed.

3.7.8 Two-speed output operation.- The design of a 1 and 36-speed synchro amplifier shall provide for the substitution of 2 and 36-speed output transmission in lieu of 1 and 36-speed output transmission. Two and 36-speed output transmission shall be supplied only when specified (see 6.2).

3.7.9 Accuracy.- The overall error of synchro amplifiers shall not exceed the values shown in table IV under all test conditions specified in 4.7.

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Table IV - Accuracy.

Condition	Error in degrees					Error in yards					
	Input speed					Input speed					
	1 and 36	1	36	2	2 and 36	1, 18 and 648	1	1 and 18	18	18 and 648	648
Static	0.1	1.0	0.1	0.5	0.1	15	3600	540	540	15	15
Dynamic (maximum velocity 32 degrees per second)1/	.1	1.0	.1	.5	.1	---	---	---	---	---	---
Dynamic (maximum velocity 6000 yards per second)1/	---	---	---	---	---	15	3600	540	540	15	15

1/ Velocity and accuracy limits are referred to the one-speed shaft.

3.7.10 Synchronization.- The synchro amplifier operation shall be self-synchronous from all displacement (including the 180-degree displacement) out of synchronism with the input synchro signal. The 1 and 36-speed and 2 and 36-speed synchro signal amplifier shall be required in no more than a total of 4.2 to 7 seconds, without external synchro loads, to achieve all of the following conditions combined, namely: to synchronize from a 180 degree displacement with the input synchro signal, to stabilize to the input synchro signal, and to attain the overall static accuracy specified in table IV for the various input speeds. For the 1, 18 and 648-speed unit, see 4.7.3.

3.7.11 Phase shift.- The equipment shall operate within the accuracy requirements of 3.7.9 for all phase shifts between 15-degree lag and 5-degree lead caused by external synchro units in the synchro transmission system.

3.7.12 Effect of synchro bus load on synchro control transformers.- In order to prevent excessive reflected errors on the synchro bus, the secondary load on the synchro control transformers shall be greater than 10,000 ohms for 60-cycle units and greater than 15,000 ohms for 400-cycle units.

3.7.13 Effect of magnetic fields.- A 5-gauss external magnetic field shall have no effect on the accuracy or the performance of the equipment.

3.7.14 Oscillations.- There shall be no spurious oscillations in the overall error as measured under any conditions covered in this specification.

3.7.15 Dial.- For the 1 and 36 or 2 and 36-speed synchro amplifiers, a dial calibrated on one face with two concentric scales differentiated hereinafter as the "outer scale" and the "inner scale" or a dial calibrated on one face with the "outer scale" and calibrated on the other face with the "inner scale" shall be provided. The outer scale shall be calibrated over its entire periphery from 0 to 360 degrees in 1-degree steps with every 5-degree mark accentuated and every 10-degree mark numbered to indicate the heading at 1 speed. The inner scale, if the dial is driven at 1 speed for 2 and 36 speed input and output operation, shall be calibrated from 0 to 60 degrees over a 60-degree portion of the dial periphery in each direction from zero. If the dial is driven at 2 speed for 2 and 36 speed input and output operation, the inner scale shall be calibrated from 0 to 60 degrees over a 120-degree portion of the dial periphery in each direction from zero. The inner scale calibration shall read from 0 to 60 degrees in 1-degree steps with every 5-degree mark accentuated and every 10-degree mark numbered. For the inner scale, calibrations to the right of the zero shall be red and marked with a (+) sign to indicate "bow down" or "starboard side up" whereas calibrations to the left of zero shall be black and marked with a (-) sign to indicate "bow up" or starboard side down". A single dial index shall be used which shall permit movement of approximately 3

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degrees in either direction for adjustment purposes. Provision shall be made to obscure that scale not in use. For the 1, 18 and 648-speed synchro amplifiers, dials shall be provided whereby the range may be read to the nearest five yards for a range of 0 - 1,296,000 yards.

3.8 Substitution of output synchro units.-

3.8.1 Provision shall be made in the design of the equipment so that any of the synchro units of corresponding size and frequency specified in table V and smaller size units, whether included in table V or not, may be substituted for the output generators specified in 1.2 without exceeding the allowable errors specified in 3.7.9.

3.8.2 The substitution of output synchros shall be accomplished by means of adapter plates and change gears. Adapter plates and change gears for synchros specified in table V shall be supplied by the manufacturer for qualification inspection specified in 4.4 when required by the Government inspector. When specified (see 6.2), a different complement of output synchros made up from those specified in 3.8.1 shall be supplied in production synchro amplifiers in lieu of the complement specified in 1.2.

Table V - Substitution of output synchro units.

Type	Speed	7G	37TX6	7DG	6DG	37TDX6	23CX4	23TX4	23CDX4	31TDX4	23TDX4
E	1	X	X	X	X	X	X	X	X	X	X
	36		X		X	X	X	X	X	X	X
	1 or 18 or 648						X		X		X
F	1	X	X	X	X	X	X	X	X	X	X
	36		X		X	X	X	X	X	X	X

3.9 The synchro signal amplifier shall meet the airborne noise requirement of MIL-I-983 for other than low-noise applications unless otherwise specified (see 6.2). The synchro amplifier shall meet the structureborne noise limits of MIL-I-983 for grade C equipment, when specified (see 6.2). There will be no structureborne noise requirement otherwise.

3.10 Drawings.- Working drawings for approval or for final submittal shall include a drawing with the data required by MIL-I-983 for preliminary drawings. For any synchro amplifier weighing 100 pounds or more the drawing shall include the weight and vertical location of the center of gravity. The final submittal of the drawing shall include the corrected weight and corrected vertical location of the center of gravity.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality control system.- The contractor shall provide and maintain a quality control system in accordance with MIL-Q-9858.

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4.3 General inspection.- The procedures for the various examination and tests to be performed on the equipment shall be as specified hereinafter. The attention of the manufacturer is invited to the applicable paragraphs of MIL-I-983 for information concerning these tests such as where conducted, failures, and test conditions.

4.4 Qualification tests[1/].- Qualification tests shall be conducted at a laboratory satisfactory to the Bureau of Ships. Qualification tests shall consist of the tests specified in table VI.

Table VI - Qualification tests.

Test	Requirement reference	Test reference
General examination	MIL-I-983	MIL-I-983
Radio frequency noise interference*	MIL-I-983	MIL-I-983
Airborne noise	MIL-I-983 and 3.9	MIL-I-983
Structureborne noise	MIL-I-983 and 3.9	MIL-I-983
Synchronization*	3.7.10	4.7.3
Enclosure	MIL-I-983	MIL-I-983
Phase shift	3.7.11	4.7.4
Effect of magnetic fields	3.7.13	4.7.5
Oscillations*	3.7.14	4.7.6
Accuracy*	3.7.9	4.7.2
Alarm operation	3.7.5	4.7.8
Input operation	3.7.7	4.7.7
Synchro connections	MIL-I-983	4.7.10
Accelerated life test	MIL-I-983	MIL-I-983
Includes vibration (4.7.12), power supply tolerance* (4.7.11), temperature and humidity (4.7.9), inclination (4.7.13), shock (4.7.14), dielectric strength*[1/] (MIL-I-983), and insulation resistance*[1/] (MIL-I-983).		

[1/] For all but rotating components.
* See 4.4.1.1(d).

4.4.1 Samples for qualification.- For qualification of individual types, the following samples will be required:

One type E - 1 and 36-speed unit (see 4.4.1.1)
One type E - 1, 18 and 648-speed unit
One type F - 1 and 36-speed unit (see 4.4.1.1)

4.4.1.1 Where qualification is desired for both type E and F, 1 and 36-speed units, one synchro signal amplifier of either type may be submitted for qualification of both types, provided the following conditions are met:

(a) The enclosure is the same for the type E and type F units.
(b) The amplifier unit can be used for either the type E or the type F units interchangeably or by using a single integral switch to convert from one type to the other.

[1/] Application for qualification tests shall be made in accordance with "Provisions Governing Qualification" (see 6.3 and 6.3.1).

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(c) The construction of the synchro unit for the type E is of the same design and materials as the synchro unit for the type F.
(d) The unit passes all the tests for one type operation and after conversion, see (b) above, passes the tests indicated with an asterisk for the other type operation.

4.5 Quality conformance inspection.- All equipment on the contract or order shall be subjected to group A and B examination and tests specified in table VII. Group C tests will be required by the bureau or agency concerned when the basic design of the equipment or when a vital part has been changed from that of the unit qualified. For example, a change from a tube type amplifier unit to a magnetic type or to a transistor type will be regarded as a basic design change. For another example, a change in gearing from 1 and 36-speed to 1 and 5-speed or from a 1 and 36-speed input and output to 1 and 36-speed input and 2 and 36-speed output will not be regarded as a basic design change. One complete equipment shall be subjected to each of the group C tests listed in table VII. Any equipment selected for group C tests shall not be used as a production equipment. Change gears and adapter plates to facilitate tests for input and output operation in accordance with 3.7.7, 3.7.8 and 3.8 shall be furnished by the contractor as requested by the Government inspector.

Table VII - Quality conformance inspection.

Inspection	Requirement reference	Test reference
Group A		
General examination	MIL-I-983	MIL-I-983
Dielectric strength 1/	MIL-I-983	MIL-I-983
Insulation resistance 1/	MIL-I-983	MIL-I-983
Accuracy	3.7.9	4.7.1
Synchro connections	MIL-I-983	4.7.10
Group B		
Oscillations	3.7.14	4.7.6
Synchronization	3.7.10	4.7.3
Alarm operation	3.7.5	4.7.8
Group C		
Radio frequency noise interference	MIL-I-983	MIL-I-983
Airborne noise	MIL-I-983 and 3.9	MIL-I-983
Structureborne noise	MIL-I-983 and 3.9	MIL-I-983
Enclosure	MIL-I-983	MIL-I-983
Phase shift	3.7.11	4.7.4
Effect of magnetic fields	3.7.13	4.7.5
Accelerated life test	MIL-I-983	MIL-I-983
Includes vibration (4.7.12), power supply tolerance* (4.7.11), temperature and humidity (4.7.9), inclination (4.7.13), shock (4.7.14), dielectric strength* 1/ (MIL-I-983), and insulation resistance* 1/ (MIL-I-983).		

1/ For all but rotating components.
* See 4.4.1.1(d).

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4.6 Rejection.- The results of each test shall be compared with specification requirements. Failure to conform to this specification for any group A, B or C test shall be counted as a defect, and the equipment shall not be offered for delivery.

4.7 Test procedures.-

4.7.1 Quality conformance inspection accuracy tests.-

4.7.1.1 Overall accuracy.- For 1 and 36-speed units, the overall accuracy of synchro amplifiers shall be determined by operating the equipment with a one and 36-speed input from a dummy director or suitable equivalent synchro input device. A type 5CT control transformer shall be provided geared to the 36-speed transmitter of the dummy director. The stator leads of this 5CT control transformer shall be connected to the 36-speed output synchro transmitter of the equipment under test, and the rotor leads of the 5CT control transformer shall be connected to a cathode ray oscillograph or other equivalent accurate voltage indicating instrument. The voltage measured across the rotor leads of the 5CT synchro control transformer represents the overall error of the equipment under test. When there is no voltage across the rotor leads of the 5CT control transformer, there is no displacement of the control transformer with respect to the output transmitter of the equipment under test, therefore no overall error exists in the synchro amplifier under this condition. Likewise, when there is a voltage indication across the rotor leads of the 5CT control transformer, this denotes a displacement of the control transformer with respect to the output transmitter of the equipment under test, therefore an error proportionate to the magnitude of the measured voltage exists in the synchro amplifier. In order to determine the error in terms of degrees and minutes for the purpose of this test, the angular deviations of the rotor of the 5CT control transformer with respect to the rotor of the synchro transmitter shall be calibrated in terms of voltage values as determined from the oscilloscope, oscillograph or equivalent device specified herein. Accuracy tests on synchro amplifiers shall be made for static and dynamic conditions in accordance with 4.7.1.1.1 and 4.7.1.1.2. When synchro differential transmitters only are required as output synchros of production synchro amplifiers the follow-up error (see 4.7.1.2) shall be substituted for the overall error. When 400-cycle synchros only are required in the output of the production synchro amplifiers the accuracy test shall be made using a 400-cycle output synchro and a 400-cycle control transformer or the 5CT as a 400-cycle synchro. Accuracy of production synchro amplifiers shall be determined using just the synchros to be delivered in the equipment. For synchro amplifiers operating at speeds different from 1 and 36 for their inputs and outputs, the test procedure shall be similar to that specified above. Each and every production synchro amplifier shall not be tested for accuracy for all of the different input and output speed variations of which the synchro amplifier is capable. For instance, a synchro amplifier with 1 and 36-speed input and output shall be tested with a 1 and 36-speed input in the manner described above and shall not be tested for a 1-speed input only, or for a 36-speed input only, or for a 2-speed input only.

4.7.1.1.1 Static accuracy.- The overall static accuracy of the equipment shall be determined by turning the dummy director slowly in not more than 10 steps and observing the maximum error. The same procedure shall be repeated in the opposite direction of rotation. The maximum overall error as indicated by the voltage across the rotor leads of the 5CT control transformer shall not exceed the value specified in 3.7.9.

4.7.1.1.2 Dynamic accuracy.- The overall dynamic accuracy shall be determined by operating the dummy director under the following conditions:

(a) Constant velocity 32 degrees per second for 32 degree per second equipments.
(b) Simple harmonic motion 40.0 degree amplitude 8.0 second period for 32 degree per second equipments.
(c) Constant velocity of 6000 yards per second for 1, 18 and 648-speed equipments.
(d) Simple harmonic motion 40.0 degree amplitude 25 second period for 1, 18 and 648-speed equipments.

For simple harmonic motion the amplitude is defined as the maximum excursion in each direction about the central heading. The dynamic error as indicated by the voltage across the rotor leads of the 5CT synchro control transformer shall not exceed the values specified in 3.7.9.

4.7.1.2 The follow-up error is defined as the error as measured (in terms of voltage converted to degrees or minutes deviation) across the rotor leads of the 36-speed input synchro control transformer in the synchro amplifier under test. This error shall be measured by means of the oscilloscope, oscillograph, or equivalent device in a manner similar to that specified in 4.7.1.1, 4.7.1.1.1 and 4.7.1.1.2.

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4.7.2 Qualification accuracy tests.- Qualification accuracy tests shall be similar to those specified in 4.7.1, with the exception that the equipment shall be tested under synchro and low inertia load for all variations of the input and output speeds of which the unit is to have the capability as specified in table IV. Adapter plates and change gears required for the installation of the various synchro units shall be provided by the contractor.

4.7.2.1 Synchro load.- The overall static and dynamic errors shall be determined using the output synchro generators and external synchro load specified in table VIII.

Table VIII - Synchro load.

Type	Output synchro generator				External synchro load			
	1-speed	18-speed	36-speed	648-speed	1-speed	18-speed	36-speed	648-speed
E	7G 31TX4	---	6G 31TX4	---	14-37TR6 2-31TR4	---	9-37TR6 2-31TR4	---
E	23TX4	23TX4	---	23TX4	2-23TR4	2-23TR4	---	2-23TR4
F	7G 31TX4	---	6G 31TX4	---	14-37TR6 2-31TR4	---	9-37TR6 2-31TR4	---

4.7.2.2 Low-inertia load.- The static and dynamic follow-up error shall be determined using the synchro units specified in table IX without external synchro load.

Table IX - Low inertia load.

Type	Synchro unit	
	1-speed	36-speed
E or F (1 and 36 speed)	37TX6 23CX4	None
E (1, 18 and 648 speed)	23TX4	None

4.7.3 Synchronization.- Tests, with external synchro loads for qualification tests and without external synchro loads for quality conformance inspection tests, shall be made to insure that the synchro amplifier will be self-synchronous as specified in 3.7.10. For this test the transmitters of the dummy director shall be set 180 degrees out of correspondence from the input synchros of the synchro signal amplifier under test. All power to the servo motor of the unit under test shall be turned off while the transmitters of the dummy director are being set. The amplifier unit of the synchro amplifier under test shall be in a ready condition (filaments heated). Then to perform the test the power to the servo motor shall be switched on and a timing device actuated. The total time it takes for the synchro amplifier to both slew 180-degrees and for the overshoots to be damped out shall be measured. The total time shall not exceed 7 seconds and the overall static accuracy of the synchro amplifier shall be within the limits of table IV. For quality conformance inspection the test shall be made with the equipment as it is to be delivered; that is, if the unit has only a 1-speed input then the test shall be made with a 1-speed input and if a 1 and 36-speed input, then the test shall be made with a 1 and 36-speed input. For qualification, the test shall be made with inputs of 1 and 36-speed or 1, 18 and 648-speed as the case may be. For the 1, 18 and 648-speed unit, there is no time limit for this test specified, since the conditions for the dynamic accuracy test are considered adequate to obtain suitable equipment. The 1, 18 and 648-speed equipment shall, however, demonstrate the capability to synchronize from 180 degrees out of correspondence with the input transmitters.

4.7.4 Phase shift.- Tests without external synchro loads on the synchro amplifier shall be made to insure that the phase shift caused by external synchros connected in the input circuit will not be

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detrimental to the operation of the synchro signal amplifier as specified in 3. [illegible] of table IV.

4.7.5 Effect of magnetic fields.- Tests without external synchro loads on the synch[illegible]ier [illegible] be made to insure that a magnetic field will not affect the accuracy of the synchro signal amp[illegible]s [illegible] ified in 3.7.13 using the input speeds required by 4.7.3 only.

4.7.6 Oscillations.- Tests shall be made to insure that oscillations are controlled as required 3.7.14.

4.7.7 Input and output operation.- Tests shall be made to insure that the operation complies with 3.7.[illegible] and 3.7.8.

4.7.8 Alarm operation.- Tests shall be conducted to insure that the alarm circuit operates as specified in 3.7.5.

4.7.9 Temperature and humidity.- The temperature and humidity test shall be conducted with the equipment operating with a constantly varying input synchro signal of simple harmonic motion of 40 degree amplitude and 8 second period. For this test the equipment shall be stack mounted or mounted between heat dissipating surfaces which will simulate a stack mounted installation. Also, the heat dissipated from the sides and back of the equipment shall be restricted by means of low heat conducting bulkheads placed adjacent to the sides and back of the equipment. The equipment shall be operated no less than 125 hours uninterruptedly at 149°F. Hot spot temperatures shall be taken in accordance with MIL-I-983 but near the end of the 149°F. test. Hot spot temperatures and temperature rise shall not exceed the limits of MIL-I-983. Any time during and after the temperature test the equipment shall be capable of meeting the static and dynamic accuracy tests of 4.7.1 or 4.7.2, as applicable.

4.7.9.1 Extended storage.- The equipment shall be tested in accordance with MIL-I-983. After stabilizing from exposures to the extreme temperature limits the equipment shall be capable of meeting the static and dynamic accuracy tests of 4.7.1 or 4.7.2, as applicable.

4.7.10 Synchro connection.- Tests shall be conducted to insure that synchro connections conform to the following:

(a) The stator and rotor leads of each synchro shall be connected to the unit terminal board and the ships wiring. (For example, connect terminal S1 of terminal board to ships wire 1MB1, R1 to ships wire 1MB, R2 to ships wire 1MBB.)
(b) An increased reading on the system (counterclockwise rotation) shall correspond with an increased reading (counterclockwise rotation) of the quantity controlling or controlled by the synchro. (For example, for a synchro that requires clockwise rotation for increasing signal, synchro terminals S1, S2 and S3 shall be connected respectivly to the unit terminal board terminals, S3, S2 and S1 corresponding to ships wire numbers 1MB3, 1MB2 and 1MB1, respectively.)

4.7.11 Power supply tolerances.- The synchro amplifier shall undergo and meet the static and dynamic accuracy tests of 4.7.1 for quality conformance inspection tests or 4.7.2 for qualification tests while being supplied power at the upper and lower limits specified in MIL-I-983.

4.7.12 Vibration test (vital equipment).- The vibration test shall be conducted in accordance with MIL-I-983. The equipment shall be operated with a constantly varying input synchro signal of simple harmonic motion of 40-degree amplitude and 8 second period for this test. During the test, the equipment shall be capable of meeting the dynamic accuracy test of 4.7.1 or 4.7.2, as applicable. After the test, the equipment shall be capable of meeting the static and dynamic accuracy tests of 4.7.1 or 4.7.2, as applicable.

4.7.13 Inclination test.- The inclination test shall be performed in accordance with MIL-I-983. The equipment shall be operated as specified in 4.7.12 and shall, during the test, be capable of meeting the dynamic accuracy test of 4.7.1 or 4.7.2, as applicable.

4.7.14 Shock test.- The shock test shall be performed in accordance with MIL-I-983. The equipment shall be operated as specified in 4.7.12 and shall after each shock be capable of meeting the

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dynamic accuracy test of 4.7.1 or 4.7.2, as applicable. After the test the equipment shall be capable of meeting the static and dynamic accuracy tests of 4.7.1 or 4.7.2, as applicable.

5. PREPARATION FOR DELIVERY

5.1 Domestic shipment and early equipment installation and for storage of onboard repair parts.-

5.1.1 Synchro signal amplifiers.-

5.1.1.1 Preservation and packaging.- Preservation and packaging shall be sufficient to afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to using activity and until early installation and may conform to the suppliers commercial practice when such meets these requirements.

5.1.1.2 Packing.- Packing shall be accomplished in a manner which will insure acceptance by common carrier and will afford protection against physical and mechanical damage during direct shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to the Uniform Freight Classification Rules and Regulations or other carrier regulations as applicable to the mode of transportation and may conform to the suppliers commercial practice when such meets these requirements.

5.1.1.3 Marking.- Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the contractor's commercial practice. The information shall include nomenclature, contract or order number, contractor's name and destination.

5.1.2 Onboard repair parts.- The onboard repair parts shall be preserved and packaged by level A; packed by level C and marked by levels A and C respectively in accordance with MIL-E-17555.

5.2 Domestic shipment and storage or overseas shipment.- Where special requirements not covered by 5.1 above, are required, the levels of preservation, packaging, packing and marking for shipment shall be specified by the procuring activity (see 6.2).

(5.2.1 The following provides various levels of protection during domestic shipment and storage or overseas shipment, which may be required when procurement is made:

5.2.1.1 Preservation and packaging, packing and marking.- The equipment and accessories, repair parts and technical publications shall be preserved and packaged by levels A or C; packed by levels A or B as specified and marked in accordance with MIL-E-17555.)

6. NOTES

6.1 Intended use.- The synchro signal amplifier in its employment in the synchro transmission system of the gyrocompass or other self-synchronous control systems is used for one or more of the following applications:

(a) To permit the operation of increased synchro loads.
(b) To permit the operation of a 400-cycle synchro load from a 60-cycle synchro bus or vice versa.
(c) To permit the transmission of a modified synchro signal.
(d) To permit the operation of a synchro load independent of the input synchro excitation.
(e) To minimize reflected errors on the input synchro signal bus.
(f) To provide a smooth and accurate transmission of synchro signals.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type required (see 1.2).
(c) Size of output synchro signal generators required (see 3.2).
(d) Inventory control point for repair parts (see 3.4).
(e) Whether unit FSN is to be entered on the identification plate.
(f) Quantity of manuals required.

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(g) When synchro transmission other than 1 and 36-speed is required (see 3.7.7 and 3.7.8).
(h) Substitute output synchros required (see 3.8.2).
(i) When grade C equipment noise limits of MIL-I-983 should be met for airborne and structure-borne noise and whether low-noise applications should be met (see 3.9).
(j) Preservation and packaging, packing and marking required if other than as specified in 5.1 (see 5.2).

6.3 With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening of bids, been tested and approved for inclusion in Qualified Products List QPL-15344 whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification, in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the qualified products list is the Bureau of Ships, Department of the Navy, Washington 25, D.C., and information pertaining to qualification of products may be obtained from that activity. Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification" (see 6.3.1).

6.3.1 Copies of "Provisions Governing Qualification" may be obtained upon application to Commanding Officer, Naval Supply Depot, 5801 Tabor Avenue, Philadelphia 20, Pennsylvania.

Preparing activity:
Navy - Ships
(Project 5990-N004(Navy))

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SPECIFICATION ANALYSIS SHEET
NAVSHIPS-4863 (8-61)

INSTRUCTIONS

BUDGET BU. NO. 45-R309

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Bureau of Ships

This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost.

Comments and the return of this form will be appreciated.

Fold on dotted lines on reverse side, staple in corner, and send to Bureau of Ships, Specifications and Standardization Branch, Washington 25, D.C.

SPECIFICATION

ORGANIZATION	CITY	STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A DIRECT GOVERNMENT CONTRACT ☐ OR A SUBCONTRACT ☐

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   a. GIVE PARAGRAPH NUMBER AND WORDING

   b. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID.

3. IS THE SPECIFICATION RESTRICTIVE? IF THE ANSWER IS "YES", IN WHAT WAY?

   ☐ YES ☐ NO

4. REMARKS (Attach any pertinent data which may be of use in improving this specification.) PLACE THIS FORM AND PAPERS IN AN ENVELOPE AND SEND TO THE BUREAU.

SUBMITTED BY (Print name and activity)	DATE

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